

MATSim User Meeting 2026

Paris, 28 September 2026



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# From Activity-Based-Model to MATSim for BRT Planning with Replan.city

## Practical Lessons from the Tel Aviv Pink Route

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Methodology and client review: Yulia Rubinshtein · Shelly Ben-Zvi

Replan.city, commissioned by Netivei Ayalon

# Netivei Ayalon: the commissioning client

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## Government-owned company

An execution arm of the Israeli Ministry of Transport

## Urban and metropolitan transport

Planning and delivery of public transport, sustainable mobility and transport innovation

Netivei Ayalon commissioned this work to support detailed BRT and feeder-network planning and flexible stop-level scenario testing.

# The Pink BRT planning task

A new BRT corridor across three municipalities in metropolitan Tel Aviv

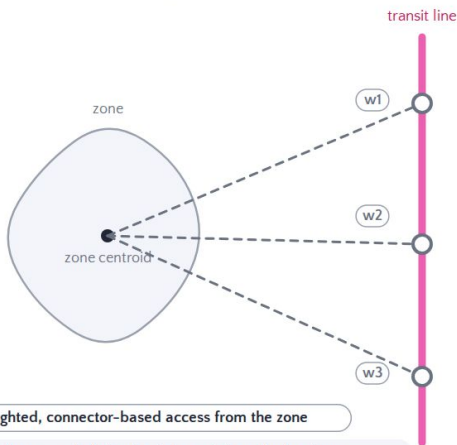


Which feeders connect to it? Where should people change? How far do they walk?

# What the planners needed to test

## ZONE-BASED ACCESS

demand enters the network from the zone through connectors



## STOP-BASED ACCESS IN THE SIMULATION

travellers walk to specific stops and board specific departures



This supports more detailed inspection of service design and passenger experience.

## 01 Stop geometry

Zonal outputs were not sufficiently granular for stop-level testing

## 02 Service dynamics

Dedicated lanes, timetables, waits and transfers interact

## 03 Iteration speed

Scenario turnaround could take several weeks

We kept the existing ABM and added a way to test changes at stop level.

# The model we assembled

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**01****2018 EMME ABM**

People, households  
and daily tours

**02****MATSim plans**

› Activities, locations  
and departure times

**03****2026 supply &  
calibration data**

› Roads, bus lanes and  
public transport  
schedules, modal split  
& traffic/pt counts

**04****Replan**

› Scenario editing,  
simulation and  
analysis, feeder  
network generation

2018 supplied the demand for calibration. 2026 supplied the network.

We kept demand fixed to compare four supply alternatives.

The study did not produce an updated 2026 demand forecast.

# Turning EMME person-days into MATSim plans

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## 1 Reconstruct

From TAZ to unique people and households

## 2 Disaggregate

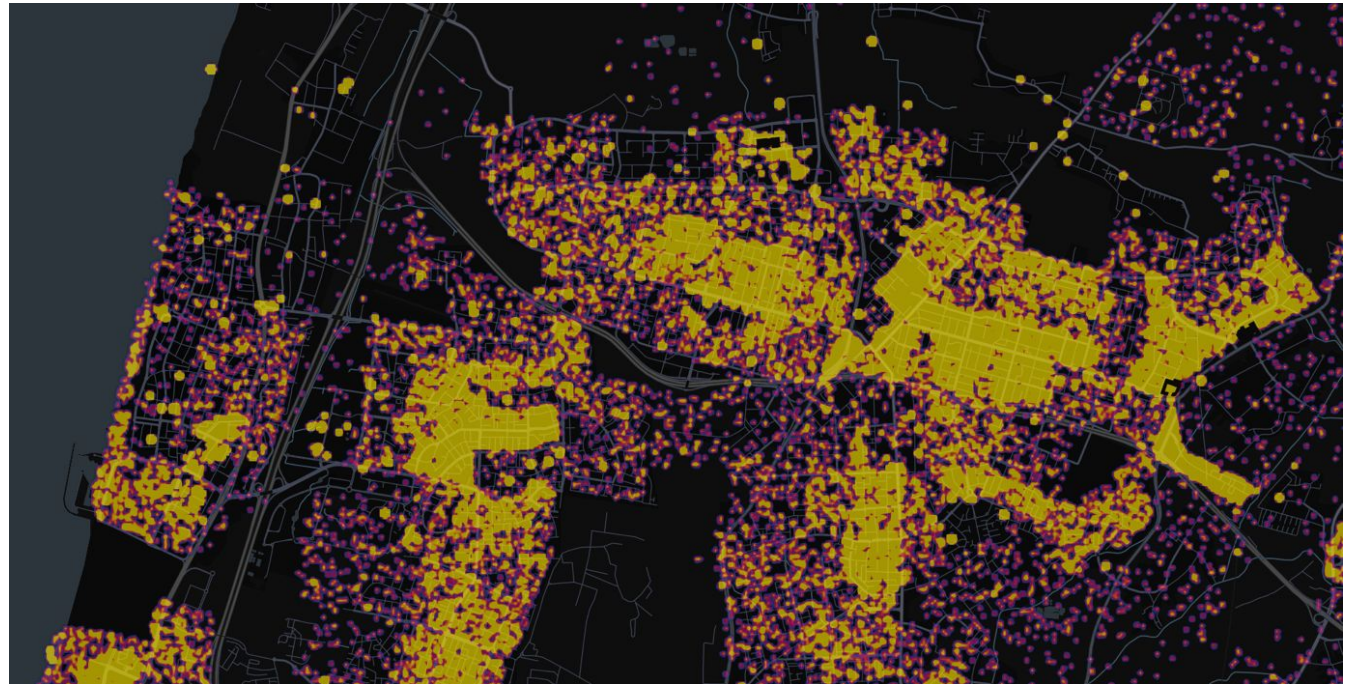
Primary and secondary tours into trip chains

## 3 Time the day

Sample departures within EMME periods using AFC profiles

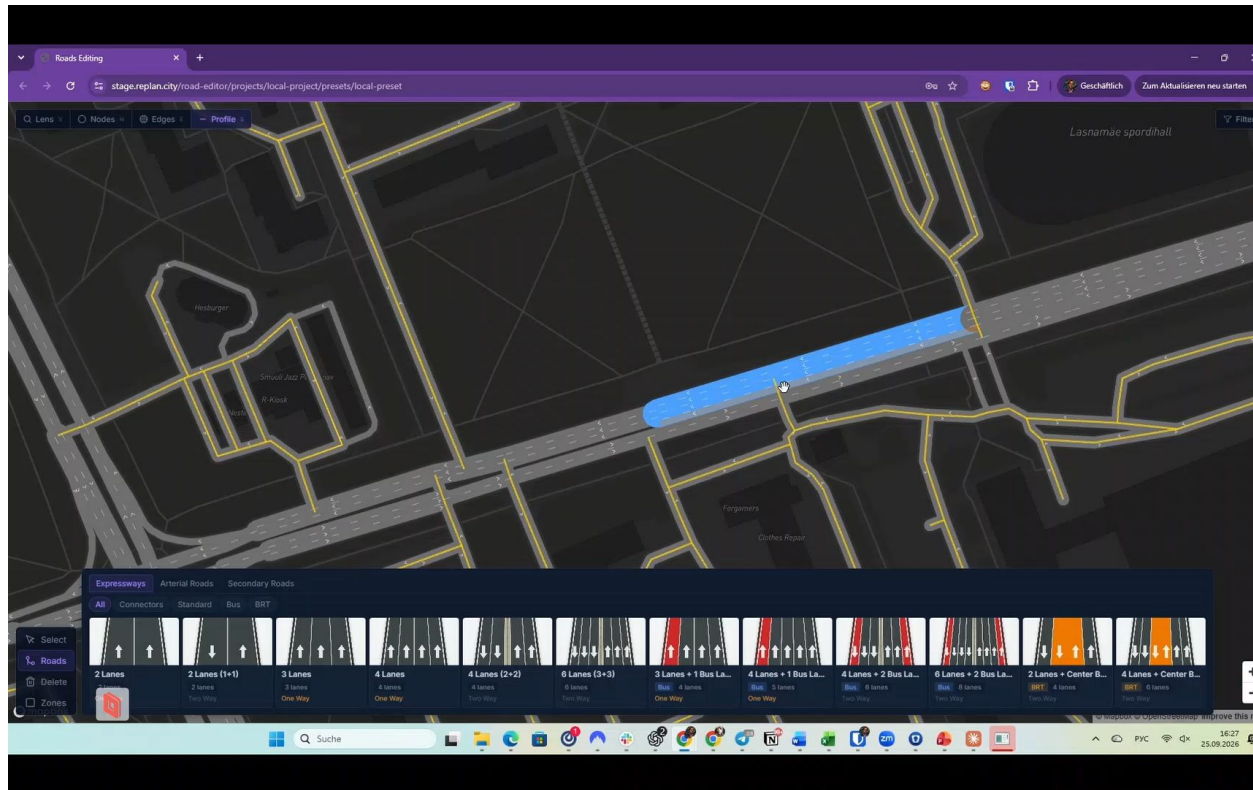
## 4 Place activities

Locations within zones, with stable home and work



Activity locations within traffic analysis zones

# Checking the physical network



Lane profiles in the Replan network editor

## Roads

Connections, direction, speed and capacity

## Public transport

Stop placement, route continuity and timetables

## Bus lanes

The count errors led us back to the lane model.

# Behaviour calibration and reference data

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| What changed          | What we checked                           | Reference                            |
|-----------------------|-------------------------------------------|--------------------------------------|
| Mode choice           | PT share and modal split                  | EMME target                          |
| Waiting and transfers | Boardings per journey, journey components | Smart-card / AFC and survey          |
| Departure times       | Time-of-day pattern                       | On-board survey as independent check |
| Line assignment       | Daily boardings by line-direction         | Smart-card / AFC counts              |

AFC timing profiles were an input. The on-board survey provided a separate check.

# How we worked through a failed check

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01

Compare

Agree the metric,  
area and reference  
data.

02

Investigate

Check the  
network, the  
journey and the  
output definition.

03

Change

Choose a  
correction and  
record the reason.

04

Run again

Compare against  
the same  
reference.

05

Review  
together

Discuss the result  
and agree the next  
test.

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The next discrepancy sent us back to the evidence.

# Calibration results and validation criteria

## Calibrated modal split

| Mode             | EMME | MATSim | Δ (pp) |
|------------------|------|--------|--------|
| Car              | 60%  | 59%    | −1     |
| Public transport | 14%  | 13%    | −1     |
| Walk             | 25%  | 28%    | +3     |

Per trip, whole simulation area. Rounded shares.

## Both contractual criteria met

PT share: −1 pp (limit ±3 pp)

91% of line-directions within ±15%  
(required ≥90%, count-fitted state)

## PT behaviour after replanning

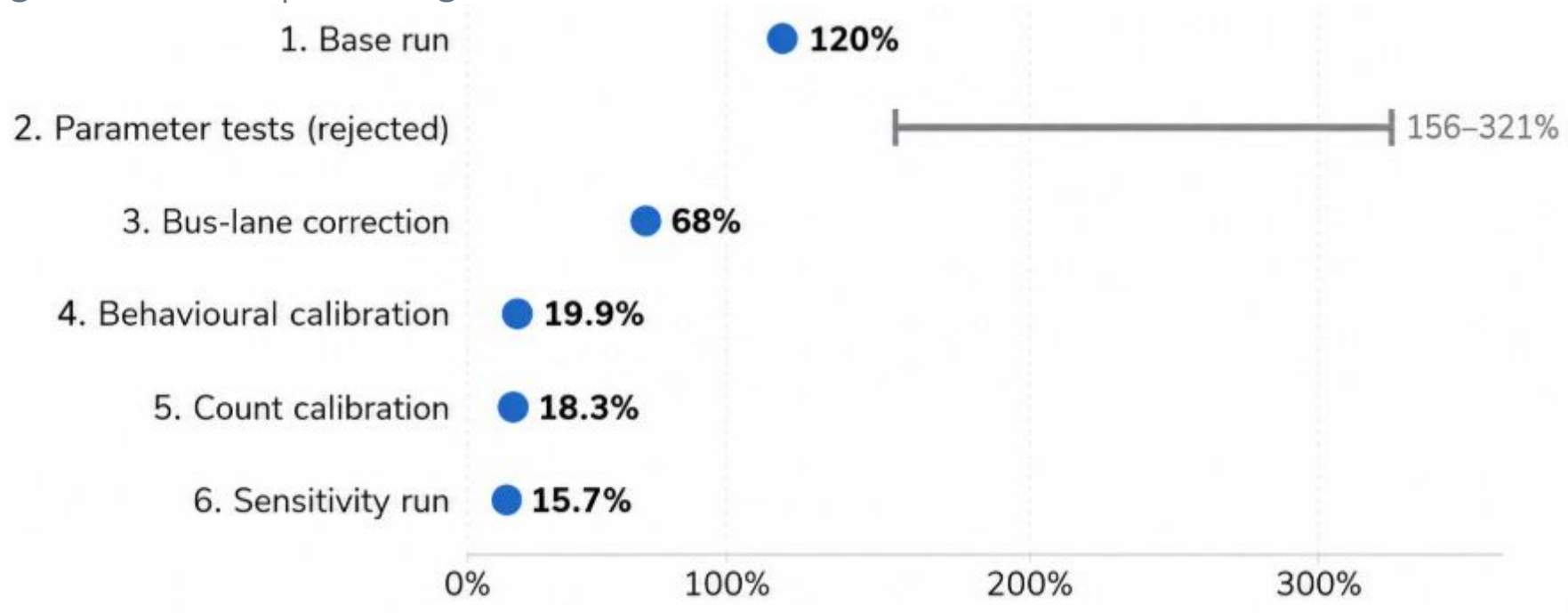
| Indicator               | Reference | MATSim  |
|-------------------------|-----------|---------|
| Waiting time            | 5.5 min   | 6.0 min |
| Boardings / journey     | 1.45      | 1.56    |
| Journey time: from area | 47.8 min  | 57 min  |
| Journey time: to area   | 45.6 min  | 55 min  |

Means. Survey times and AFC boardings.  
Journey times: target area, door to door, unfiltered.

Full journey times remain 19–21% above the survey.

# How the line-count fit changed after calibration stages

Passenger-weighted absolute percentage error (WAPEE)



The sensitivity run started with the count-adjusted population.

**Count adjustment added 1.6 percentage points of improvement.**

# Why were some journeys so long?

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Following individual journeys helped explain the averages

**Waiting for a particular service**

A missed route variant could mean a long wait.

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**Queues and walking access**

Bus queues and access/egress assumptions affected journey times.

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**The output definition**

One extracted “in-vehicle” measure also included waiting.

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We investigated the tails with the client. Some limitations remained.

# What changed in the acceptance review?

| Conventional static four-step                                             | MATSim in this project                                                                                                |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Zonal demand and assigned flows<br>Check trips, mode shares and volumes.  | The same aggregates, plus complete journeys<br>Reconcile tours, trips and boarded legs. Inspect timing and transfers. |
| A calibrated demand matrix<br>Document changes to demand and the network. | A calibrated population and plan state<br>Record which plans changed and what agents may still replan.                |
| Fit for the planning task<br>Check sensitivity and agreed tolerances.     | Evidence at several levels<br>Review journeys, lines and corridors. Separate contractual tests from diagnostics.      |

**Several rounds of client review challenged the benchmarks, outliers and response to change.**

Comparison with aggregate/static practice. The project's EMME demand was already activity-based.

# We should have explained the outlier sooner

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## A large error on one feeder raised a much bigger question

### What the reviewer saw

One feeder carried far more passengers in the model than in the observations.

The reviewer asked whether this could also inflate BRT demand.

### What our investigation added

Passengers shifted between parallel feeders.

Combined demand at the shared hubs was much closer to the observations.

The feeder was not capacity-constrained.

The mechanism belonged beside the number. The local uncertainty still belonged in the report.

**Replan's mission coming to life:**

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**Transit planners (operational and strategic) without  
PhD from TU Berlin or ETH Zürich working  
with MATSim model**

# Editing a planning alternative

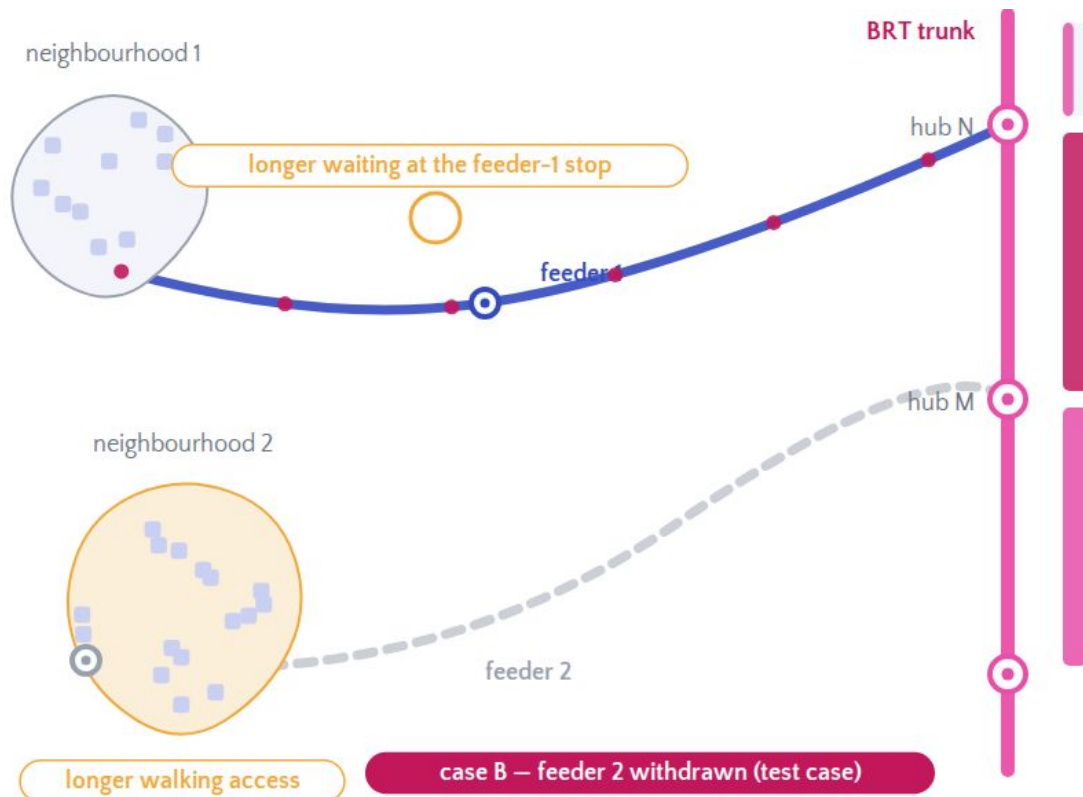


Illustration: removing a feeder changes access and transfers

A small edit can take minutes. The simulation runs separately.

## Sketch a feeder

Draw a route and choose its stops.

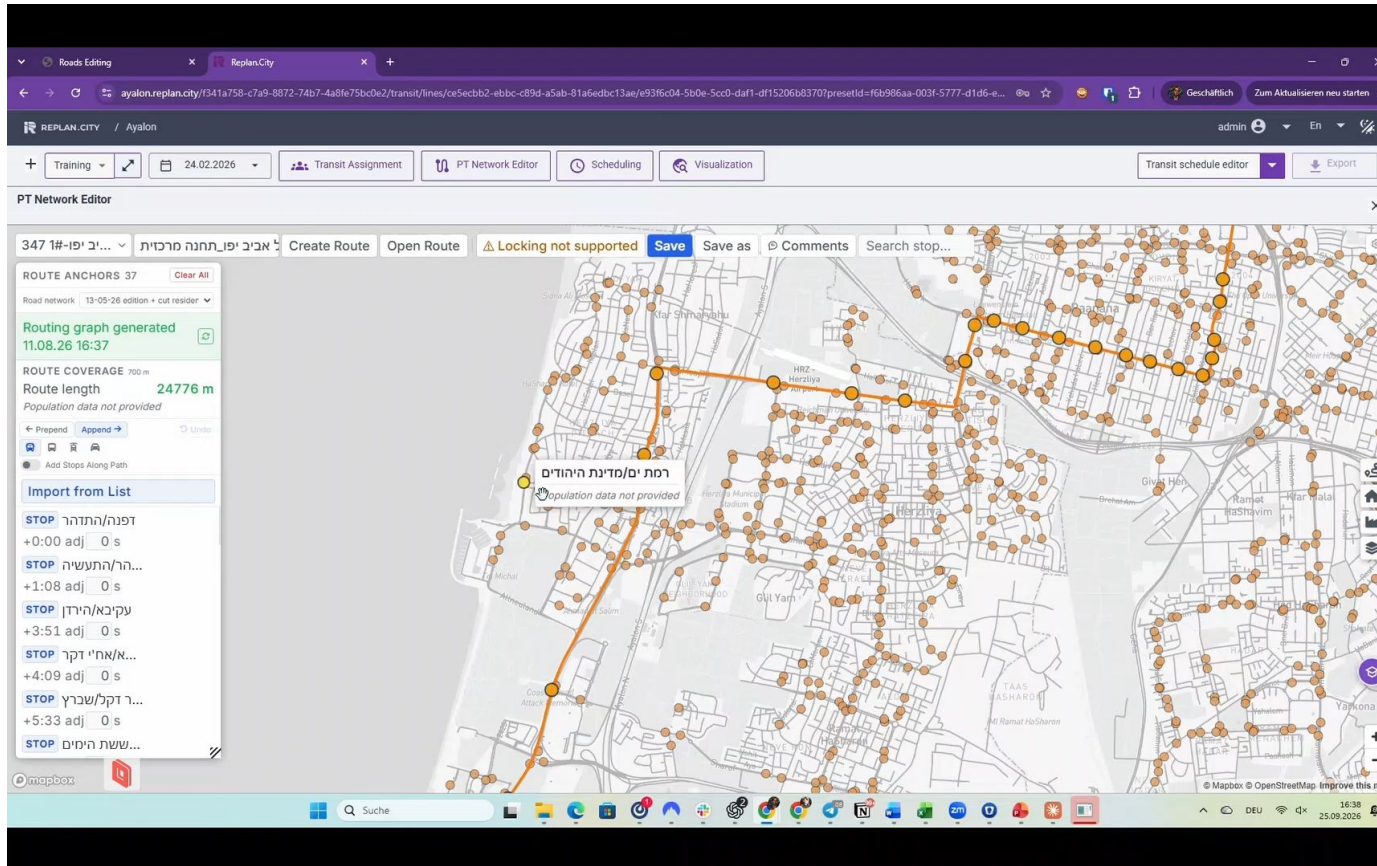
## Change the BRT

Remove a stop or change the alignment.

## Save and run

Keep the alternative separate from the baseline.

# Editing a planning alternative



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Draw a route and choose its stops.

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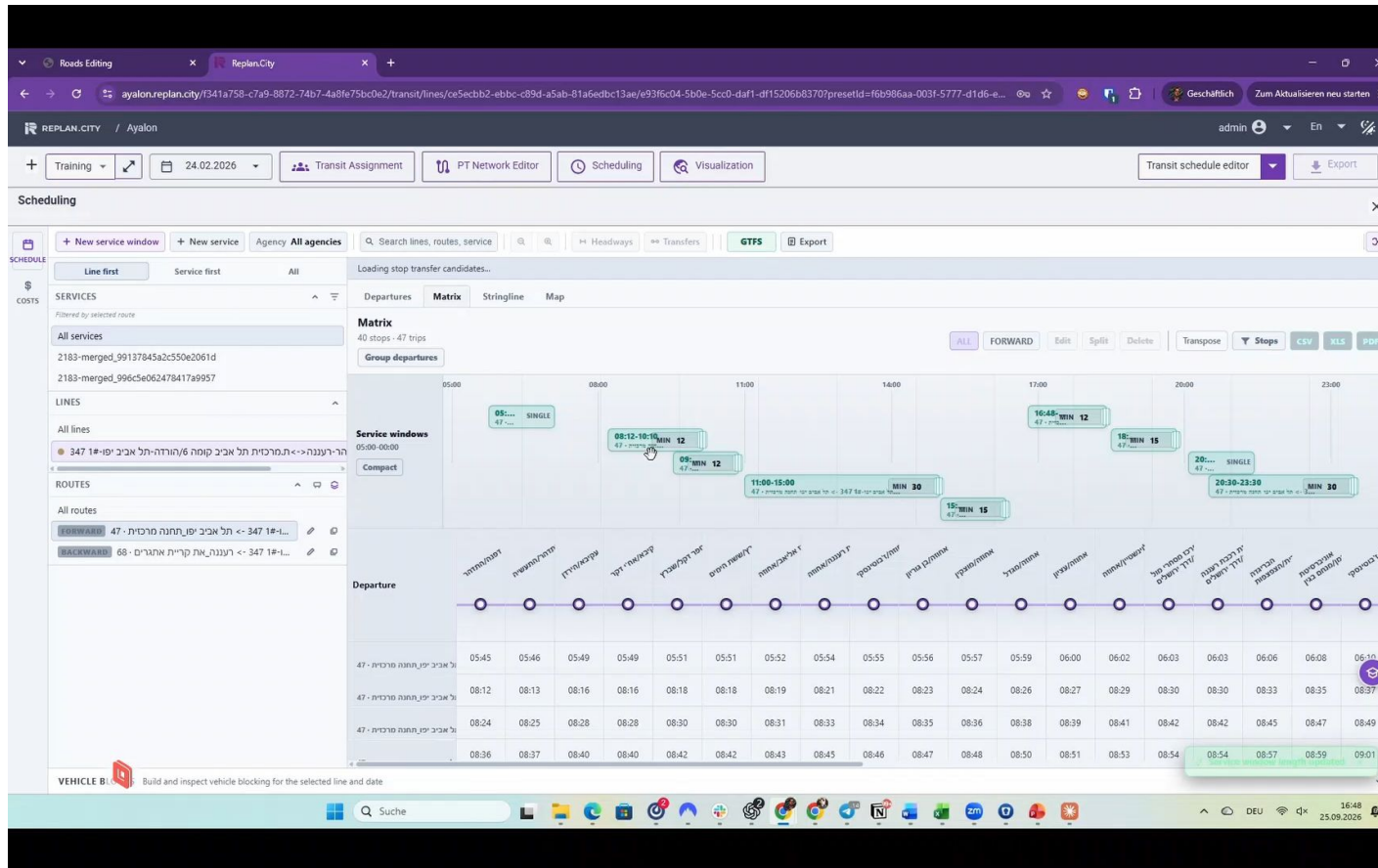
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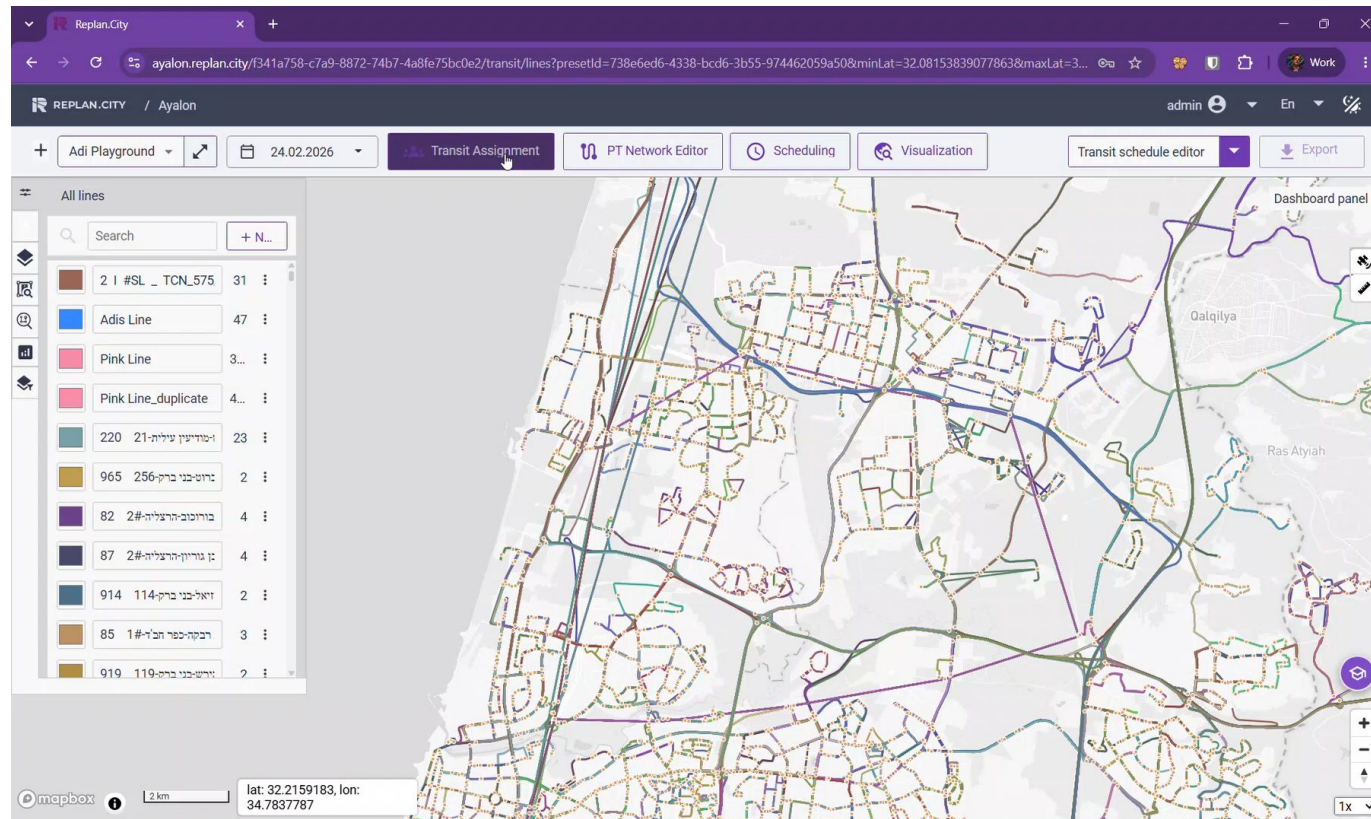
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# Feeder network: A generated network still needs an engineer



Candidate network from the study

## The generator

Kept fixed services and excluded unsuitable roads while proposing feeder routes.

## The engineer

Checked coverage and connections, removed duplication and set the timetable.

**MATSim then tested the refined alternative.**

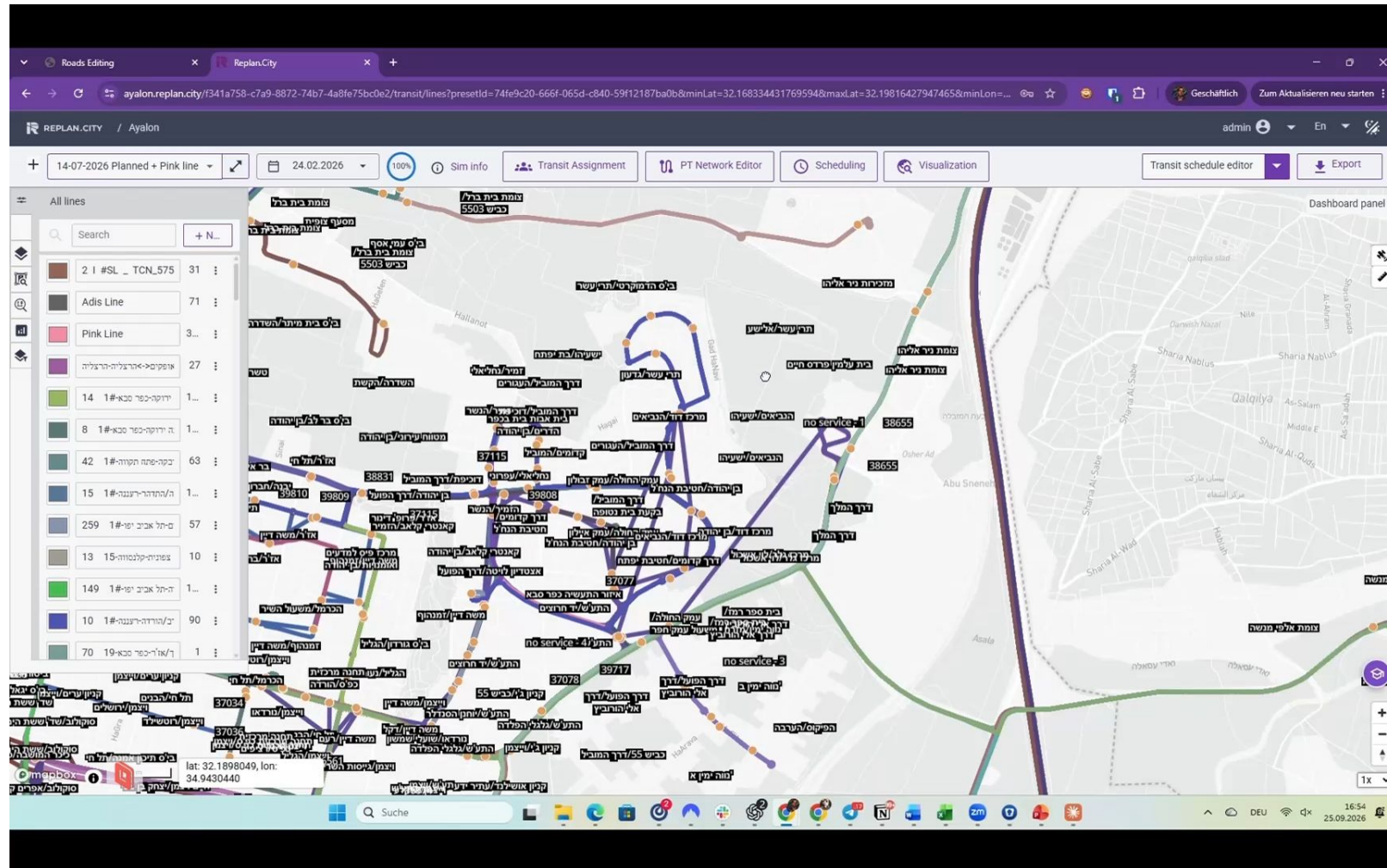
# Checking & comparing accessibility in Replan

The screenshot displays the REPLAN.CITY web application interface. The top navigation bar includes the REPLAN.CITY logo, a user profile (admin), language settings (En), and a 100% zoom level. Below the navigation bar, there are tabs for '14-07-2026 Planned + Pink line', '24.02.2026', 'Sim info', 'Transit Assignment', 'PT Network Editor', 'Scheduling', and 'Visualization'. A 'Transit schedule editor' dropdown and an 'Export' button are also present.

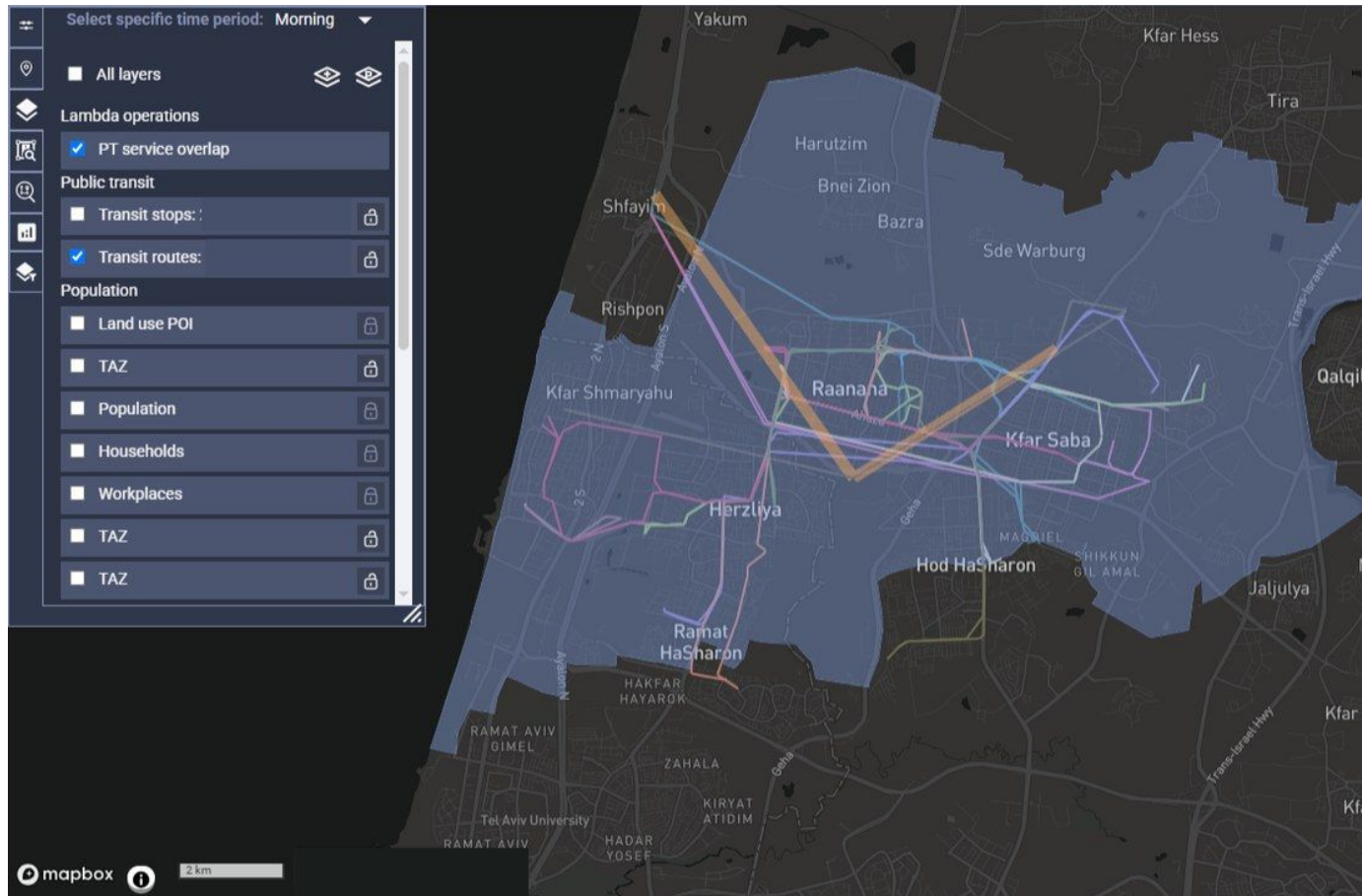
The left sidebar contains the 'Analysis type' section with four options: 'Find Optimal Route', 'Destination Accessibility' (selected), 'Activity Accessibility', and 'Travel Efficiency'. Below this, the 'Parameters' section for 'Destination Accessibility' includes a 'Zone' field, a 'From point' field with coordinates (34.941811, 31.961769), a 'Start time' field (09:00:00), and a 'Compare with other preset' checkbox. A 'Start calculation' button is located at the bottom of the parameters section. The 'Result' section shows 'Destination Accessibility' and a 'Run a calculation to see results' button.

The main map area displays a network of roads and transit lines. A red area highlights the 'From point' location. The map includes labels for various locations such as Nablus, Balata RC, Beit Furik, Barta, Aqraba, Duma, Adey Ad, Turmus Ayya, Abwein, Salit, Bedia, Shuqba, Qibya, Bidrus, Ramallah, Biddi, Jericho, Aqbat Jaber, Almog, and Kalya. The map also shows natural features like 'Harama Nature Reserve' and 'Nahal Prat Nature Reserve'. A 'Features distribution' panel is visible at the bottom right.

# Analyzing individual lines and stops after the simulation



# Learning the full workflow in Replan



Network and GIS layers in Replan

## One complete planning task

Start with the agreed baseline.

Make a service change and run it.

Use dashboards, isochrones and GIS to explain the difference.

# What we would carry into the next project

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- 1 Explain surprising results before handing them over**  
Put the mechanism and its planning consequence beside the number.
- 2 Review each calibration stage with the client**  
Agree what changed and what the evidence supports as the work proceeds.
- 3 Test the response to a changed network**  
A good baseline count fit alone does not establish scenario validity.
- 4 Teach a complete planning task & Set clear boundaries for simulations**  
The handover includes editing, running and explaining a comparison.

Thank you to the Replan and Netivei Ayalon teams

# Thank you!

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